# A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-22304666})$, $p = 5$

Generated by `tools/certificate_guide.gp`

## What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic $L_x/K$, the normalized generator $\sigma_x$, the pair $(a', J)$ representing the torsion class $e_j$, the auxiliary divisor $I'$, the compactly represented element $t$, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1-\sigma_x)^2 I' + \mathrm{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

## Notation and conventions

**Ideals as matrices.** An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 1025 & 922 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over $K$ with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,1025 + \mathbf{Z}\,(922 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

**Characters.** The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character $x$ on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number $j$ records which basis element $e_j$ of $\mathrm{Cl}(K)[5]$ the stored pair $(a', J)$ represents.

**Automorphisms.** $\sigma_x$ is a field automorphism of $L_x$. Writing $\theta$ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of $L_x$.

## Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 22304666 = 0$, of discriminant $-89218664$. Class group invariants $(310 \quad 5 \quad 5)$, class number 7750; the three stored generator ideals (HNF) are $\begin{pmatrix} 237 & 164 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by $-1$ of order 2.

## Entry 1: character $a$, column $e_1$

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - y^4 + (s - 575)\, y^3 + (24s + 97717)\, y^2 + (-43s + 1659874)\, y + (-1614s + 1807464)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1149*y^8 + 196584*y^7 + 25759605*y^6 + 958544598*y^5
+ 18565428401*y^4 + 204280939244*y^3 + 1421675858334*y^2 + 909630185
8536*y + 61370491822632
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(-1 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + (\frac{3904}{1131408212890625})\, s, \qquad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2011971826836 = 2^2 \cdot 3^8 \cdot 19 \cdot 59 \cdot 68389$:

$$\begin{pmatrix}
4139859726 & 343395000 & 2535244326 & 2063698974 & 382998456 & 3977215220 & 2673539728 & 3540337911 & 4022145501 & 2141703746 \\
0 & 3 & 0 & 0 & 0 & 0 & 1 & 2 & 0 & 1 \\
0 & 0 & 9 & 6 & 6 & 2 & 3 & 8 & 1 & 5 \\
0 & 0 & 0 & 3 & 0 & 0 & 2 & 1 & 2 & 1 \\
0 & 0 & 0 & 0 & 6 & 3 & 1 & 1 & 3 & 5 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 165 factors with signed exponents of absolute value up to 4641384731; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 2: character $a$, column $e_2$

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - y^4 + (s - 575)\, y^3 + (24s + 97717)\, y^2 + (-43s + 1659874)\, y + (-1614s + 1807464)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1149*y^8 + 196584*y^7 + 25759605*y^6 + 958544598*y^5
+ 18565428401*y^4 + 204280939244*y^3 + 1421675858334*y^2 + 909630185
8536*y + 61370491822632
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(-1 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + (\frac{8377}{8202010025025000})\, s, \qquad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3086462423376 = 2^4 \cdot 3^3 \cdot 19 \cdot 199 \cdot 1889603$:

$$\begin{pmatrix}
128602600974 & 96613396386 & 43447505518 & 91124890311 & 5974728758 & 101959981973 & 110677499833 & 68379935548 & 48100140050 & 124877916841 \\
0 & 6 & 2 & 5 & 2 & 5 & 3 & 1 & 3 & 5 \\
0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 877492637; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 3: character $a$, column $e_3$

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - y^4 + (s - 575)\, y^3 + (24s + 97717)\, y^2 + (-43s + 1659874)\, y + (-1614s + 1807464)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1149*y^8 + 196584*y^7 + 25759605*y^6 + 958544598*y^5
+ 18565428401*y^4 + 204280939244*y^3 + 1421675858334*y^2 + 909630185
8536*y + 61370491822632
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(-1 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625}\right) s, \qquad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $157110630918021 = 3^6 \cdot 11 \cdot 83 \cdot 236051773$:

$$\begin{pmatrix}
1939637418741 & 1259274219294 & 671565227709 & 1124033623668 & 313731578856 & 1101481365564 & 235373477572 & 791704982717 & 690060676963 & 1931537858284 \\
0 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\
0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 3 & 1 & 0 & 0 & 1 & 0 & 2 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 3 & 2 & 0 & 2 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 170 factors with signed exponents of absolute value up to 4176178902; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 4: character $b$, column $e_1$

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 2y^4 + (-s - 1283)\, y^3 + (-49s + 56092)\, y^2 + (-756s + 3363456)\, y + (-1548s + 34778840)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2562*y^8 + 117316*y^7 + 30453299*y^6 + 2098029052*y^5
+ 81654727066*y^4 + 2009646785008*y^3 + 31346092288336*y^2 + 286159
962069696*y + 1263016472100064
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(1 \quad 0 \quad 0 \quad 1 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + (\frac{3904}{1131408212890625})\, s, \qquad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $37321958351972 = 2^2 \cdot 7^4 \cdot 163 \cdot 23841011$:

$$\begin{pmatrix}
380836309714 & 349127883664 & 326166156547 & 300503090777 & 260035030516 & 352731983919 & 54382164384 & 221182113499 & 87349334833 & 286488993511 \\
0 & 14 & 7 & 0 & 6 & 7 & 13 & 4 & 8 & 7 \\
0 & 0 & 7 & 1 & 1 & 0 & 0 & 0 & 6 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 241 factors with signed exponents of absolute value up to 27682657; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 5: character $b$, column $e_2$

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 2y^4 + (-s - 1283)\, y^3 + (-49s + 56092)\, y^2 + (-756s + 3363456)\, y + (-1548s + 34778840)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2562*y^8 + 117316*y^7 + 30453299*y^6 + 2098029052*y^5
+ 81654727066*y^4 + 2009646785008*y^3 + 31346092288336*y^2 + 286159
962069696*y + 1263016472100064
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(1 \quad 0 \quad 0 \quad 1 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + (\frac{8377}{8202010025025000})\, s, \qquad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $689321696 = 2^5 \cdot 7 \cdot 113^2 \cdot 241$:

$$\begin{pmatrix} 762524 & 28928 & 743978 & 208950 & 199895 & 283 & 374462 & 522642 & 118486 & 277796 \\ 0 & 226 & 112 & 114 & 113 & 198 & 44 & 0 & 141 & 207 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 237 factors with signed exponents of absolute value up to 23507574; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 6: character $b$, column $e_3$

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 2y^4 + (-s - 1283)\,y^3 + (-49s + 56092)\,y^2 + (-756s + 3363456)\,y + (-1548s + 34778840)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2562*y^8 + 117316*y^7 + 30453299*y^6 + 2098029052*y^5
+ 81654727066*y^4 + 2009646785008*y^3 + 31346092288336*y^2 + 286159
962069696*y + 1263016472100064
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(1 \quad 0 \quad 0 \quad 1 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625}\right) s, \qquad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $29923882795576 = 2^3 \cdot 7^2 \cdot 11 \cdot 6939675973$:

$$\begin{pmatrix} 14961941397788 & 2721761786549 & 4331885870012 & 4996643103853 & 6154131505386 & 4291710260372 & 5299884086854 & 12874011541561 & 12150013129842 & 6491287023458 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 234 factors with signed exponents of absolute value up to 39777566; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 7: character $c$, column $e_1$

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - y^4 - 3276y^3 + (32s + 163370)\,y^2 + (-1628s - 5034956)\,y + (21360s + 73909920)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 6551*y^8 + 333292*y^7 + 335524*y^6 - 912510488*y^5 +
82370946756*y^4 - 4453347079152*y^3 + 139107409683120*y^2 - 22955148
70041600*y + 15639171215040000
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right) s, \qquad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $17333214462000 = 2^4 \cdot 3^4 \cdot 5^3 \cdot 73 \cdot 79 \cdot 18553$:

$$\begin{pmatrix} 96295635900 & 61876954230 & 76593294616 & 46930014924 & 96004604488 & 54109049711 & 39315928898 & 76760808089 & 96249509450 & 92463269999 \\ 0 & 15 & 1 & 12 & 2 & 7 & 7 & 7 & 4 & 10 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 168 factors with signed exponents of absolute value up to 1299592122154; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 8: character $c$, column $e_2$

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - y^4 - 3276y^3 + (32s + 163370)\, y^2 + (-1628s - 5034956)\, y + (21360s + 73909920)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 6551*y^8 + 333292*y^7 + 335524*y^6 - 912510488*y^5 +
82370946756*y^4 - 4453347079152*y^3 + 139107409683120*y^2 - 22955148
70041600*y + 15639171215040000
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right) s, \qquad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25751690626500 = 2^2 \cdot 3^6 \cdot 5^3 \cdot 89 \cdot 793813$:

$$\begin{pmatrix} 19075326390 & 705655890 & 3266055590 & 8336355333 & 8347812428 & 18750583840 & 18771048831 & 14287340942 & 4065395727 & 5743682280 \\ 0 & 45 & 0 & 35 & 44 & 24 & 7 & 10 & 21 & 24 \\ 0 & 0 & 10 & 7 & 2 & 8 & 5 & 6 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 168 factors with signed exponents of absolute value up to 2777918197332; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 3 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 9: character $c$, column $e_3$

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - y^4 - 3276y^3 + (32s + 163370)\, y^2 + (-1628s - 5034956)\, y + (21360s + 73909920)$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 6551*y^8 + 333292*y^7 + 335524*y^6 - 912510488*y^5 +
82370946756*y^4 - 4453347079152*y^3 + 139107409683120*y^2 - 22955148
70041600*y + 15639171215040000
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625}\right) s, \qquad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $237047164054650 = 2 \cdot 3^2 \cdot 5^2 \cdot 67 \cdot 79 \cdot 7547 \cdot 13187$:

$$\begin{pmatrix}
47409432810930 & 18226675286445 & 19396256919290 & 26564155288417 & 5719458227677 & 14868344071665 & 30874904452391 & 20179779199322 & 28277677276695 & 8799753669678 \\
0 & 5 & 2 & 1 & 3 & 0 & 1 & 1 & 1 & 3 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 2706310439783; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 10: character $a + b$, column $e_1$

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 + 4220y^3 + 10sy^2 + 4868019y + 20350s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 + 8440*y^8 + 27544438*y^6 + 43316546960*y^4 + 32775608046361*y^
2 + 9236864045585000
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(2 \quad -1 \quad -1 \quad -2 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right) s, \qquad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $32380840522000 = 2^4 \cdot 5^3 \cdot 349 \cdot 1697 \cdot 27337$:

$$\begin{pmatrix}
809521013050 & 636914682420 & 703215958438 & 398336920708 & 269685070550 & 705101951840 & 239535646061 & 120960869397 & 82083845502 & 661064393056 \\
0 & 10 & 6 & 4 & 8 & 8 & 7 & 8 & 2 & 6 \\
0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 200 factors with signed exponents of absolute value up to 10369746757; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 11: character $a+b$, column $e_2$

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 + 4220y^3 + 10sy^2 + 4868019y + 20350s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 + 8440*y^8 + 27544438*y^6 + 43316546960*y^4 + 32775608046361*y^
2 + 9236864045585000
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(2 \quad -1 \quad -1 \quad -2 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right) s, \qquad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $105422785037020 = 2^2 \cdot 5 \cdot 11 \cdot 37^2 \cdot 79 \cdot 367 \cdot 12073$:

$$\begin{pmatrix}
1424632230230 & 628032865510 & 422113434189 & 196354615504 & 349002126102 & 1346453271136 & 1004849654733 & 263059059484 & 1177517468987 & 312535138748 \\
0 & 74 & 1 & 73 & 0 & 46 & 53 & 0 & 0 & 68 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 196 factors with signed exponents of absolute value up to 4307192759; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 12: character $a+b$, column $e_3$

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 + 4220y^3 + 10sy^2 + 4868019y + 20350s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 + 8440*y^8 + 27544438*y^6 + 43316546960*y^4 + 32775608046361*y^
2 + 9236864045585000
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(2 \quad -1 \quad -1 \quad -2 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625}\right) s, \qquad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $381290360000 = 2^6 \cdot 5^4 \cdot 11^2 \cdot 78779$:

$$\begin{pmatrix}
86656900 & 46118270 & 51138190 & 67208238 & 6309924 & 23466316 & 78826356 & 55931577 & 10144121 & 4154391 \\
0 & 110 & 50 & 24 & 84 & 92 & 42 & 86 & 61 & 31 \\
0 & 0 & 10 & 8 & 8 & 9 & 9 & 3 & 4 & 3 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 202 factors with signed exponents of absolute value up to 181731065; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 13: character $a + c$, column $e_1$

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 1052y^3 + 182212y - 576s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2104*y^8 + 1471128*y^6 - 383374048*y^4 + 33201212944*y^2 + 74
00152866816
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + (\frac{3904}{1131408212890625})\, s, \qquad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2008703366028 = 2^2 \cdot 3^5 \cdot 97 \cdot 21304817$:

$$\begin{pmatrix} 12399403494 & 4050750762 & 5705174976 & 1999300920 & 9282945174 & 8234810009 & 11543884875 & 8933328743 & 11592209811 & 11541840 \\ 0 & 3 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 6 & 3 & 0 & 5 & 0 & 4 & 4 & 3 \\ 0 & 0 & 0 & 3 & 0 & 2 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 275 factors with signed exponents of absolute value up to 28486885; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 14: character $a + c$, column $e_2$

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 1052y^3 + 182212y - 576s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2104*y^8 + 1471128*y^6 - 383374048*y^4 + 33201212944*y^2 + 74
00152866816
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + (\frac{8377}{8202010025025000})\, s, \qquad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9950729965776 = 2^4 \cdot 3^6 \cdot 97 \cdot 8794997$:

$$\begin{pmatrix}
5118688254 & 2052748482 & 1760408034 & 4938845682 & 1543787364 & 4619095182 & 2937414356 & 2024979838 & 2408132052 & 688575088 \\
0 & 6 & 0 & 0 & 3 & 0 & 5 & 5 & 3 & 5 \\
0 & 0 & 3 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\
0 & 0 & 0 & 3 & 0 & 0 & 2 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 6 & 0 & 3 & 4 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element $t$ is stored in compact form as a product of 264 factors with signed exponents of absolute value up to 72158913; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 15: character $a + c$, column $e_3$

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 1052y^3 + 182212y - 576s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 2104*y^8 + 1471128*y^6 - 383374048*y^4 + 33201212944*y^2 + 74
00152866816
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + (\frac{28414}{124636871337890625})\, s, \qquad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $474945511098312 = 2^3 \cdot 3^2 \cdot 6596465431921$:

$$\begin{pmatrix} 39578792591526 & 779469330886 & 19504140858906 & 9498948272704 & 13057321657732 & 37059068354727 & 10298056479453 & 6709420054922 & 2214858801416 & 34003988981637 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 268 factors with signed exponents of absolute value up to 167281467; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 16: character $b + c$, column $e_1$

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 2352y^3 + 24sy^2 + 1606412y + 32s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 4704*y^8 + 8744728*y^6 + 5290925568*y^4 + 2614819480720*y^2 +
22839977984
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right) s, \qquad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6131132579636 = 2^2 \cdot 797 \cdot 33343 \cdot 57679$:

$$\begin{pmatrix} 3065566289818 & 1503637013046 & 755449518008 & 518193045176 & 2732433849554 & 2519667887477 & 2652942676250 & 284925175767 & 116629298813 & 2093749454496 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 547 factors with signed exponents of absolute value up to 2253329443234709431183837; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 17: character $b + c$, column $e_2$

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 2352y^3 + 24sy^2 + 1606412y + 32s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 4704*y^8 + 8744728*y^6 + 5290925568*y^4 + 2614819480720*y^2 +
22839977984
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right) s, \qquad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12579753651008 = 2^6 \cdot 196558650797$:

$$\begin{pmatrix} 786234603188 & 156664761530 & 34451188536 & 181332555294 & 785572568024 & 202059719132 & 246618447494 & 401012949447 & 178928714585 & 387125885272 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 545 factors with signed exponents of absolute value up to 1300856544693427787457525; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

## Entry 18: character $b + c$, column $e_3$

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of $x$ on the classes of the three stored generators of $\mathrm{Cl}(K)$.

### The extension and its normalized generator

Relative polynomial of $L_x$ over $K$:

$$y^5 - 2352y^3 + 24sy^2 + 1606412y + 32s$$

Absolute polynomial of $L_x$ over $\mathbf{Q}$ (degree 10):

```
y^10 - 4704*y^8 + 8744728*y^6 + 5290925568*y^4 + 2614819480720*y^2 +
22839977984
```

The automorphism $\sigma_x$ is determined by the image of the primitive element $\theta$: the coordinates of $\sigma_x(\theta)$ in the integral basis of $L_x$ are $(0 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625}\right) s, \qquad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

### The auxiliary pair $(t, I')$

$I'$ is the ideal of $L_x$ with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $433874004813712 = 2^4 \cdot 31 \cdot 874745977447$:

$$\begin{pmatrix} 108468501203428 & 34716882907440 & 31889153178230 & 72488941472930 & 105641363862040 & 15965878080521 & 62553761915726 & 85085457697953 & 16658379344833 & 8394878856656 \\ 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element $t$ is stored in compact form as a product of 547 factors with signed exponents of absolute value up to 2243372457587227807225125; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.